DC-DC Buck Converter Report B2-G5

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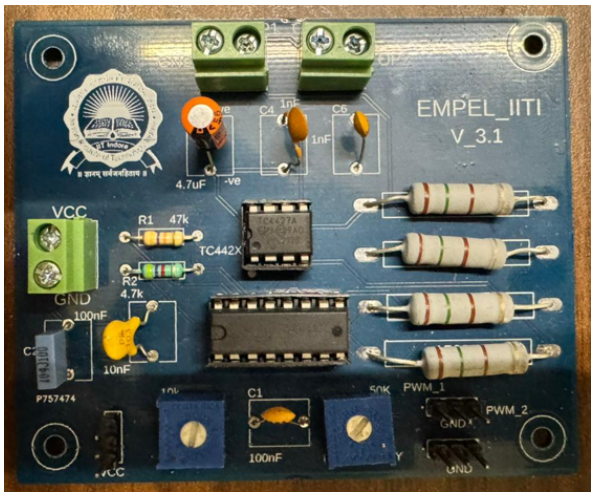
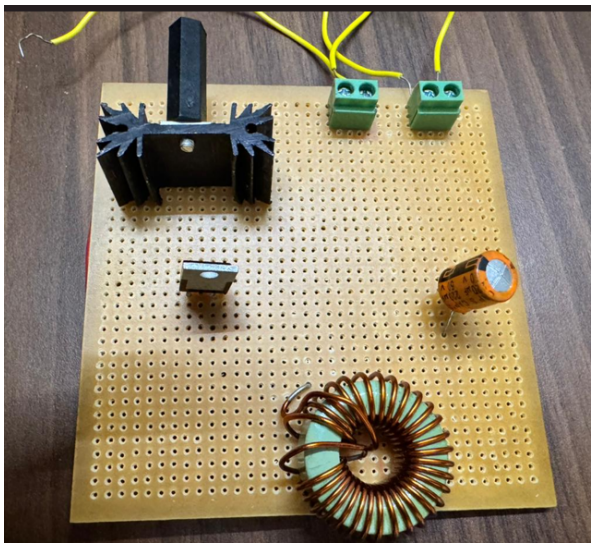
Objective: Design and analyze a DC-DC Buck Converter using TL494 and observe CCM and DCM modes.
Specifications: Input 20 V, Output 7.5 V, Switching frequency 20 kHz, Output current 1 A
Drive Link: `https://drive.google.com/file/d/1GbvmQU773Ghvoxdu7S2UesZ6Kj_O1g1M/view?usp=drive_link`

Hardware

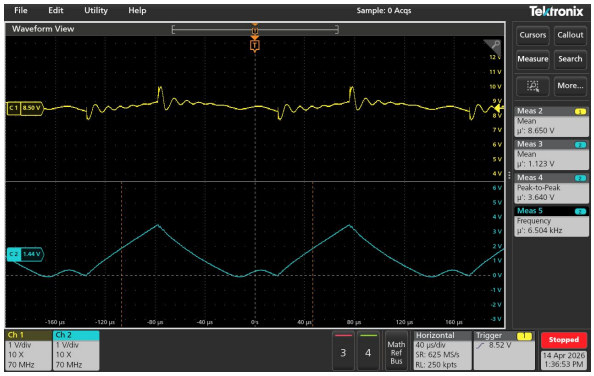


(a) Gate Driver



(b) Converter

CCM



(a) Output current and voltage

(b) Inductor current w.r.t gate pulse

DCM

Inductor current reaches zero

Conclusion: The Output Voltage of Buck Converter is given by $V_o = DV_{in}$.